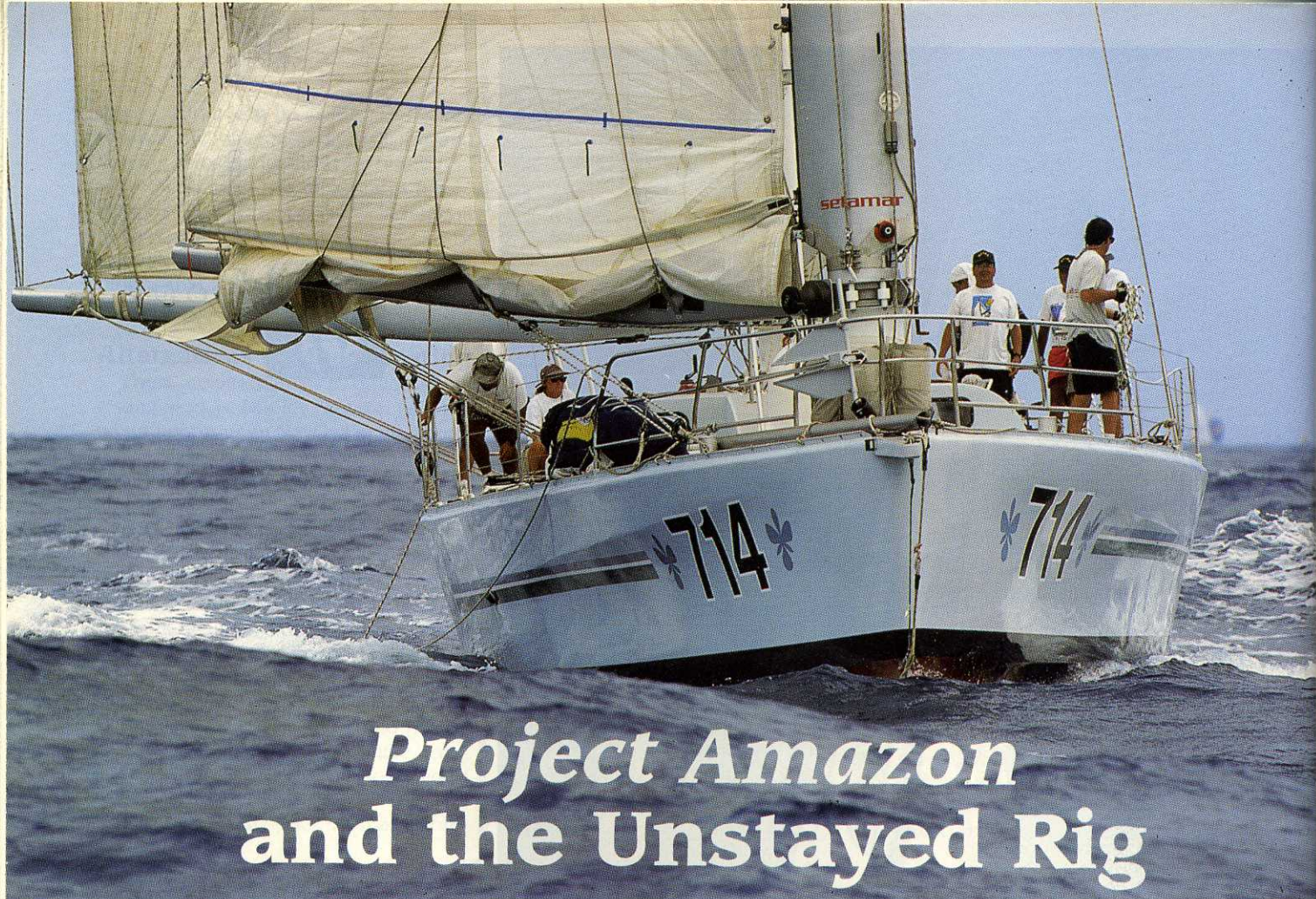




Project Amazon and the Unstayed Rig

DAN NERNEY

An atypical Open 60-class aluminum cat-ketch designed for the grueling Around Alone race is fitted with an unstayed rig. And proponents of wingmast setups are watching closely.

by Eric Sponberg

On September 26, an Open 60 built to my design, *Project Amazon*, embarks on the Around Alone Singlehanded Race (formerly called the BOC Challenge), with owner-skipper Sebastian Reidl (rhymes with “bridle”) at the helm. *Project Amazon* is a cat-ketch, equipped with two carbon fiber free-standing wingmasts; that is, the masts are shaped like wings, rotate to follow the wind, and have no wires holding them up. They carry a fore and a mizzen, as well as headsails, staysails, and an asymmetrical spinnaker.

The only other boat ever to compete in an around-the-world event with a rig in any way similar to *Project Amazon*'s was *Lady Pepperell*, Tony Lush's much-modified 54' Hunter that he sailed in the 1982–83 premier running of the BOC. She was a cat-ketch, and her masts (round-sectioned, not shaped like wings) rotated to roll up the sails for easy reefing and stowage. Unfortunately, Lush was forced

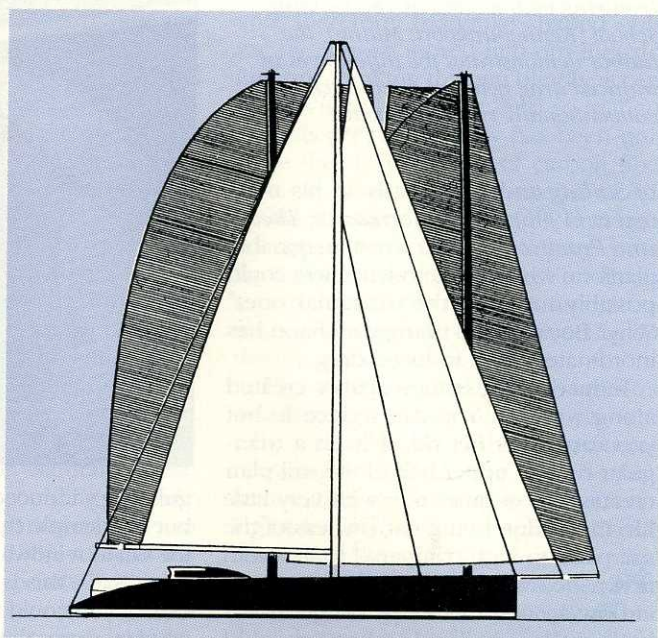
to abandon *Lady Pepperell* on the second leg in the Indian Ocean when she lost her keel. His being plucked off the boat by fellow competitor Francis Stokes literally minutes before *Lady Pepperell* sank was one of the most spectacular events of that race.

I've been active in the design and development of free-standing rigs since 1980, when I was chief engineer at what was then Tillotson-Pearson Inc., builder of Garry Hoyt's Freedom Yachts. In that year, TPI switched the building of Freedom masts from aluminum to carbon fiber, and I did the original engineering for the new carbon fiber spars. (Freedom Yachts now builds its own boats in Middletown, Rhode Island, and Goetz Marine Technology in nearby Bristol builds their masts.) I left TPI at the end of 1981, and in my career have designed over 50 free-standing masts.

From time to time, I and others, most notably Garry Hoyt, have appeared in

print and in person to speak about the sublime simplicity, safety, and improved aerodynamics of free-standing rigs—often to skeptical listeners. Their reaction is always the same: sailors simply don't trust a sailboat mast to stand all by itself without any wires holding it up. Never mind that these same skeptics will board a 747 airliner that weighs 800,000 lbs and flies at 500 mph seven miles above the Earth—with no wires holding the wings on! Well, no one ever claimed that sailors are logical beings.

Although the virtues of free-standing rigs are compelling, people are not spurred to buy new-and-improved boats, and I am not really surprised by that. It's like computers—just because the latest and greatest computer is 10 times faster than mine doesn't mean I'm going to spend \$4,000+ right this minute to upgrade. My computer works fine for what I do, and I'll upgrade when I am forced to do so by my clients or my competitors, con-



ERIC SPONBERG

Facing page—Project Amazon undergoes sea trials before owner/skipper Sebastian Reidl races her singlehanded around the world. **Left**—By using a pair of free-standing, rotating wing masts, Reidl hopes to capitalize on the predominantly reaching-oriented legs of the long race. **Above**—The cat-ketch rig (shaded) has more sail area and, argues the author, more efficiently shaped sails than a conventionally stayed rig (unshaded).

sistent with the balance in my check-book. The same is true with sailboats—they all work quite fine as they are, and upgrading to a newer model is not a spontaneous decision. “What really needs to happen,” says Garry Hoyt, “is for free-standing rigs to be legitimized on the race course.” *Project Amazon* in the Around Alone Race represents the best opportunity I’ve had to prove out such a rig in a major racing event.

How *Project Amazon* Came to Be

In 1992, Reidl saw an article I wrote for a German yachting magazine about my Globetrotter cat-ketch designs—two singlehanded racers and two family cruisers. He wrote to me, explaining that he once owned a machine shop in Vancouver, British Columbia, in which he had built an aluminum sailboat that he had raced shorthanded in the Pacific. Now living in Ecuador and out of competitive sailing for about 10 years, he wanted to race again, this time singlehanded around the world, in an aluminum boat.

To win the race, first you have to finish, and over the years many of the top-notch composite boats in the BOC and its companion race, the Vendée Globe (nonstop around the world single-

handed; Around Alone has four stops), came to grief against ice, submerged objects, and rocks. Aluminum is pretty indestructible compared to composites—you can hit an iceberg at speed and only dent the hull—so Reidl’s chances of finishing would improve as a result.

The problem to solve was speed. In general, for the same sail area, lighter-weight composite boats are faster than heavier aluminum boats. The Globetrotter designs convinced Reidl that he could hang a lot more sail area from two masts than he could from one, which would compensate for the heavier weight of an aluminum hull (see **drawing above**).

In addition, he would solve another prime cause of not finishing: dismasting. The modern history of round-the-world racing is replete with tales of boats losing their rigs over the side. A typical 60’ round-the-world racer with one three-spreader mast has over 500 separate parts holding it up, most of which are at heights where you cannot inspect them easily. If any one part fails or falls out—a cotter pin, say, or clevis pin, or shackle—there is probably a 95% chance that the rig will fall down. The boat is out of the race. By contrast, on a free-standing rig, there are no wires and fittings. If it is not there, it cannot fail. Only two structural

points hold up each mast—the deck partners and the maststep. Four parts total for *Project Amazon*. These are substantial structures located where you can see them all the time. The safety and reliability of the rig go up in proportion, 500+ to 4. That’s at least 12,500%!

Boats crash and get dismantled in every race. But here is a rig simple enough, reliable enough, and powerful enough to get a man in his late 50s around the world in a comparatively heavy boat and still be competitive against the young, virile (but crazy) French whippersnappers in their relatively fragile rocket ships.

Free-standing vs. Stayed Rigs

For a given mast height, the cat-ketch rig can set not only more sail area than a conventionally rigged sailboat, but more efficient sail area when the masts are shaped like wings and can rotate into the wind. The extra efficiency comes from two sources: reducing induced drag in the sail plan, and rotating the mast.

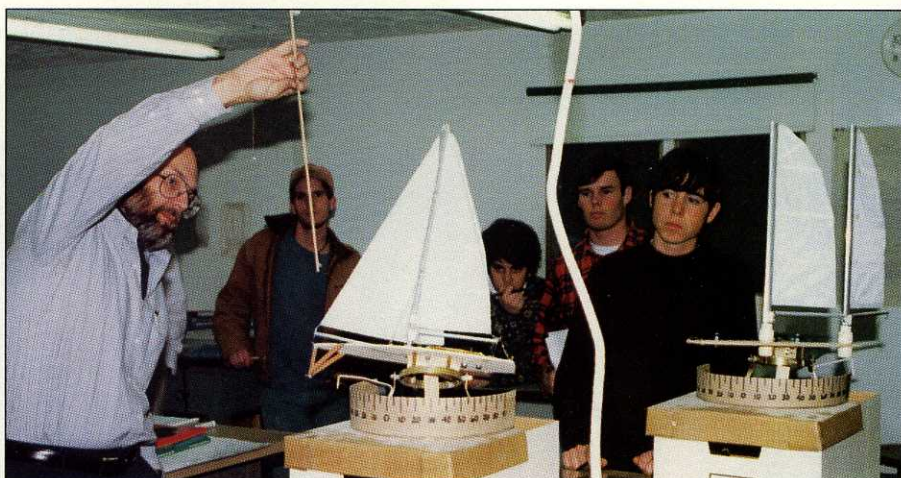
Sails set on free-standing masts are not confined to a triangular shape by the presence of the headstay and backstay (see **drawing above**). Aerodynamicist C.A. Marchaj, author of the most definitive modern books on the science of sailing, states both in his *Aero-Hydrodynamics*

Lecturing with models at The Landing School (Kennebunkport, Maine), the author demonstrates the differences in induced drag between a cat-ketch and conventionally rigged sailboats.

of Sailing and particularly in his most recent (1996) *Sail Performance, Theory and Practice*, that "the worst imaginable planform which the aerodynamicist could possibly invent is the triangular one." Why? Because the triangular shape has inordinately high induced drag.

Induced drag is inescapably created along with lift. You can reduce it, but you can never get rid of it. In a triangular rig, the upper half of the sail plan creates almost pure drag with very little lift; this is due to the narrowness of the sail plan up high compared to its width down below, along with the resulting airflow around this shape. In guest lectures, such as at The Landing School of Boatbuilding and Design (Kennebunkport, Maine), I have demonstrated the differences in lift and drag between a triangular stayed rig and a cat-ketch wingmast rig using some small gimbaled models and a fan (see **photo, above right**).

Theoretically, you can reduce a trian-



ERIC SPONBERG

gular rig's induced drag if you could do but one simple thing—twist the head of the sail to windward. Unfortunately, with fabric sails, this is impossible to do, and even if you could, twisting a sail to windward increases the angle of attack awfully close to stall, where lift suddenly drops to zero. It would be a continual knife-edge balance between power and stall, which cannot be done when you're sailing through waves—the rig moves around too much. So if we insist on wires to hold up the mast, we must accept a more

or less triangular sail plan, which is just not very powerful.

If free-standing rigs are so good, why do designers, builders, and sailors insist on wires and, therefore, triangular sails? After all, we would not be caught dead in boats with triangular keels and rudders, which are lifting surfaces just like sails; we'd be laughed right off the water. We know the science below the waterline, and design new keel and rudder shapes every day, yet we blatantly disregard the self-same science above the

Free-standing Rigs in Review

I date the modern era of free-standing rigs in larger sailboats from Garry Hoyt's founding of Freedom Yachts in 1976. Paul Petronello, president of Freedom Yachts in Middletown, Rhode Island, remembers that when he started sailboat racing in the 1960s, the major rig controversy was whether aluminum spars should be allowed to replace their wooden counterparts. At the time, aluminum had been available for over 30 years, but it took that long to convince people that aluminum was a smart design-and-engineering choice. Carbon fiber has also been available now for over 30 years, and in free-standing rigs generally since 1980. Since then, a number of production and custom yachts with free-standing rigs, most made of carbon fiber, have logged significant racing records and sailing passages. Others are notable for their design. Herewith is a small selection from my files, as well as contributions from friends and colleagues.

—Eric Sponberg



PATRICK MOULIGNE

FROG KISS

Freedom 44

Owner: Patrick Mouligne, Portsmouth, Rhode Island.

One of the winningest Freedom yachts ever. Mouligne has raced *Frog Kiss* in every Newport-Bermuda-1-2 (singlehanded out, doublehanded back) race since 1989, logging 12 first-place finishes in class and fleet in each leg, and winning it outright in 1995. Mouligne also won the 1987 Newport-Bermuda-1-2 outright in his Freedom 32 *Sloop du Jour*. With his family, Mouligne sailed *Frog Kiss* to Venezuela and back in 1996-97.



RON JOHNSON

SEA PEARL 28

Marine Concepts Inc.

The Sea Pearl 28 and its smaller sister, the Sea Pearl 21, are trailerable cat-ketches that make a good first boat in which to experience the world of free-standing rigs. Their carbon fiber free-standing masts were designed by Eric Sponberg. Marine Concepts president and Sea Pearl designer Ron Johnson says the advantages of the rig are many: "One person can raise and lower the masts; the masts flex with the gusts to spill the wind, rather than making the boat heel suddenly; as the masts straighten out, power pulls the boat forward; the masts rotate, giving better off-wind performance and lessening the danger of an uncontrolled jibe, with a safety zone of steering over 180°; you can luff up when running straight downwind, keeping the stern to the wind; heave to by sheeting the aft sail and releasing the forward sail, and the boat weathervanes unattended." Johnson believes that "when the public starts seeing the advantages, you will witness the decline of the stayed rig."

waterline; marconi-sloop sail plans have remained basically the same for over a hundred years. The reason we do this has nothing to do with science or engineering; it is purely political—design and handicap rating rules for racing sailboats require that wires hold up the rig! And rationally or no, cruising sailboats follow the developments of racing sailboats, because cruising sailboat builders do not want to cut off half their market by preventing a boat from racing should a potential customer want to.

In the predominant handicapping rule of today, the International Measurement System (IMS), rule 203, paragraph 7, states, "...a yacht must be fitted with a bona-fide forestay." This is a direct carry-over from the International Offshore Rule (IOR) of the 1960s and '70s, which stated exactly the same thing in rule 802.6A. The history of rigging requirements in rating rules can be traced back for decades.

The forces on the rig and the engineering balance of a stayed-mast structure require that if you have a forestay, you have to have a backstay. And if you have stays, then you must also have shrouds—the loading on the rig demands it. Therefore, all standing rigging wires

are required by the totally arbitrary and political rules that govern racing sailboat design, and standing rigging gives us predominantly triangular sails.

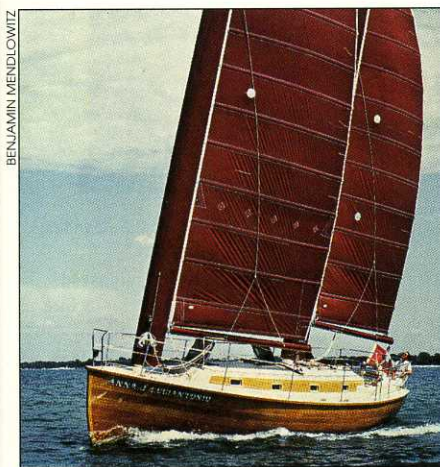
The reasons for this development go back thousands of years. Somewhere deep in history, someone discovered that they could hold up taller and taller masts by a collection of ropes, which was lighter, cheaper, and easier than trying to find ever bigger trees—the most convenient and naturally tall things around. As rigs got taller, ships got bigger and carried more cargo at cheaper rates, shipping companies made more money, and the ship owners could afford to build and race boats for pleasure. Naturally, these boats were built with standing rigging.

By the time sailboat rating rules began measuring sail area in 1883, some of the natural measurement points in the rig were where standing rigging attached to the hull, deck, and mast. Standing rigging was taken for granted. While many evolutionary changes have occurred in rig design over the years—most notably in new materials, first with metals, then with composites—standing rigging still remains bound inside the rating rules with no relief in sight. Wires in the rig,

and, therefore, triangular-shaped sails, are by now so inbred into our industry and our thinking that we blindly accept them without question. And so triangular sails prevail. It really has been only since the development of carbon fiber that we finally have a material that is strong enough, stiff enough, and light enough to make free-standing rigs practical. And there is no question that a stayed carbon fiber mast is as light as you can design a rig today, if you accept the encumbrances of rigging.

The second feature that increases the efficiency of a free-standing rig, and which equally falls victim to traditional rating rules, is a rotating mast. So far I have traced the prohibition against rotating masts back to yacht designer L. Francis Herreshoff, who had a patent on a rotating mast design; he installed one on an R-class boat (20' LWL) in 1925. But when the regatta committee of the New York Yacht Club heard about this rig, it promptly passed the rule prohibiting "revolving masts, double luffed sails, and similar contrivances." This prohibition remains in place. IMS rule 203.6 states, in its entirety: "Rotating masts are excluded for yachts measured under the Rule." Nor

Continues on page 56



ANNA J. LUGIANTONIO

Halsey Herreshoff 35

Owner: Paul Mellon, Sarnia, Ontario, Canada.

Built in 1980, *Anna J. Lugiantonio* was one of the first monohulls to have a free-standing wingmast rig, which was built in wood and epoxy by Gougeon Brothers (Bay City, Michigan). Mellon sails the Great Lakes, cruising as far north as Lake Huron's Georgian Bay. He blows past C&C 35s and Alberg 37s regularly, and literally sails circles around a local Freedom 40. At anchor, Mellon turns the masts 80° to each other; this stops the boat dead in the water, preventing it from sailing around its mooring.



MOLLY B

Sparhawk 42

Owner: Jeff Strong, Deerfield Beach, Florida.

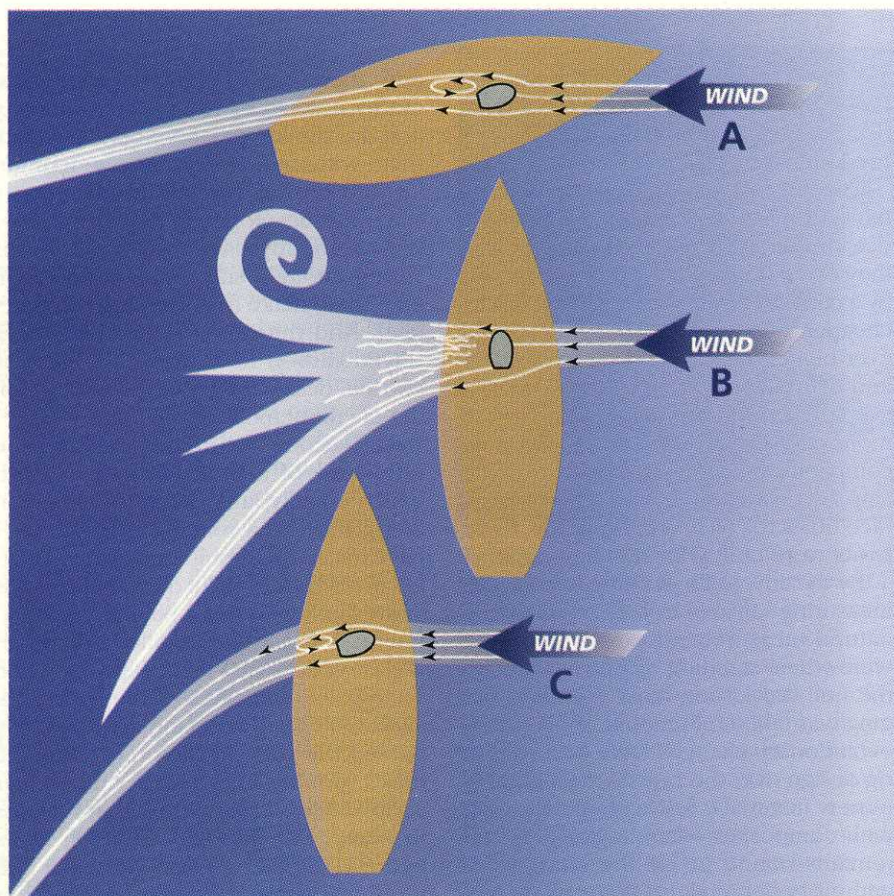
Molly B won first in PHRF Class 7 in the 1991 Key West Race. A broken boom vang at the start caused her to start behind Class 8. The vang broke again overnight, and one of the four crew members was seasick the entire race—even so, *Molly B* finished more than half an hour ahead of the Class 7 second-place finisher, a Cal 40. The Sparhawk 42 and its sister, the Sparhawk 36, are Richard Black designs and were built by Cat Ketch Yachts Inc. (Port Arthur, Texas), which also built the Halsey Herreshoff 28', 31', 38', and 45' cat-ketches. All Sparhawk and Herreshoff cat-ketches have Sponberg-designed carbon fiber spars.



NONSUCH 30

Hinterhoeller Yachts

The Nonsuches came in a series of five designs at 22', 26', 30', 33', and 36', of which about 1,200 were built. The early Nonsuch 30 pictured here, designed by Mark Ellis, was equipped with a Gougeon-built wood-and-epoxy wingmast, which later proved to have too much projected area. The boat was subsequently retrofitted with the standard aluminum free-standing mast, and this unit has stood up well on all the Nonsuches. Carbon fiber masts are available as retrofits from Composite Engineering, builders of *Project Amazon's* masts. Intended for freshwater and coastal cruising, Nonsuches nonetheless have sailed across the Atlantic and Pacific oceans. The Nonsuches sell quickly on the used market, thanks to their good, common-sense design and a strong owners' association comprised of over 600 boats, the largest yacht owners' class in the world.



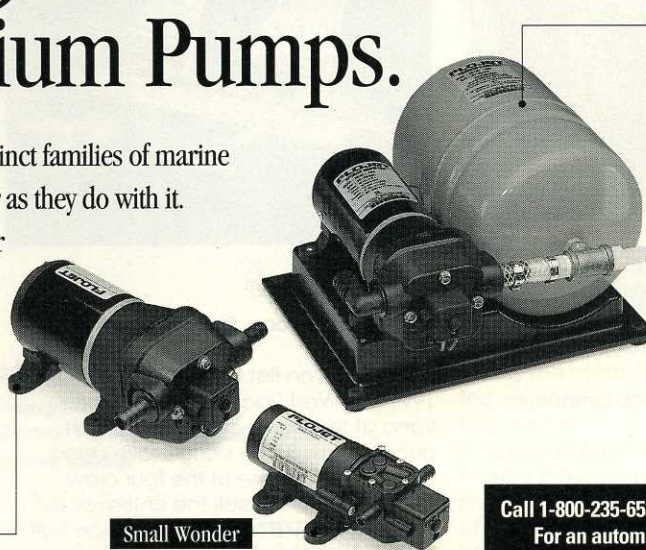
When sailing on the wind (A), the airflow on the back side of the mast is well attached. Off the wind (B), the flow separates easily on the other side of a non-rotating mast, producing almost pure drag. By allowing a mast to rotate into the wind (C) the airflow remains attached and creates considerable lift.

are there any rule changes in sight.

Despite the rules, rotating a stayed mast is difficult, to say the least, because the rigging wires simply get in the way. Sailing on the wind, the airflow attaches fairly well over the mast and mainsail (see **diagram left**). Off the wind, though, the mainsail sets to the side of the mast, and its leading-edge shape—the most important part of the rig for generating power—is terrible. The airflow quickly separates off the mast. To recover some lift, the sailmaker has to build enough camber into the sail to fool the airflow into reattaching. Most people will say that the more downwind you go, the more pure drag you want anyway, to push you downwind. They obviously have not felt the adrenaline rush of acceleration and speed caused by pure lift from a properly designed and rotated wingmast rig. Get rid of the wires, and full mast rotation is possible. Rotate the

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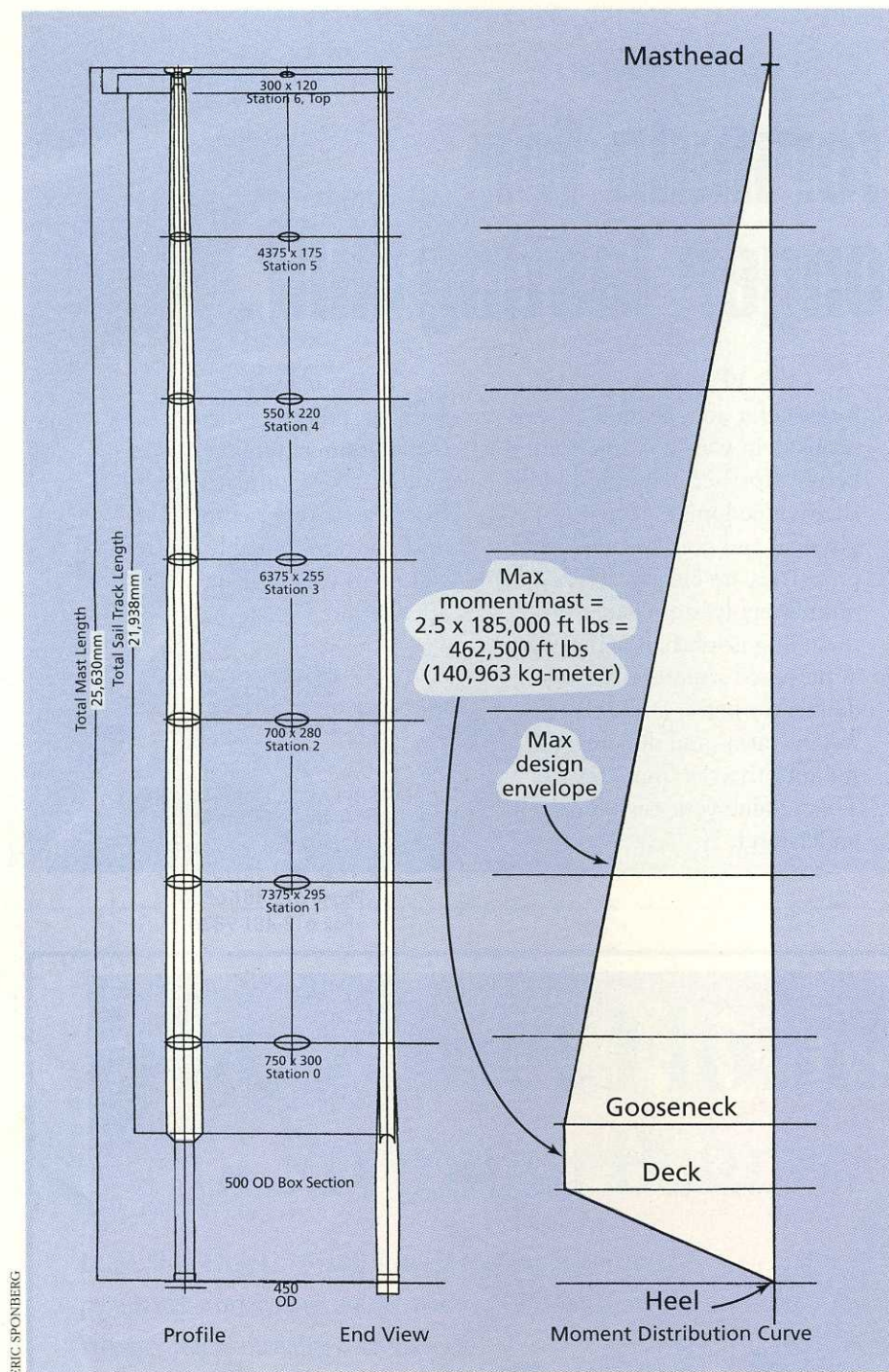
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mast, and all the aerodynamics change. Even downwind—especially downwind—pure lift is much more powerful than pure drag. The boat is considerably faster because so much more power is harnessed from the wind.

Another benefit of eliminating the wires, and probably the most important for a shorthanded long-distance race, is that the boat is much more directionally stable and more resistant to jibing. A boat with a stayed rig can sail perhaps 10° by the lee before the mainsail jibes. If not properly controlled, the boom will swing violently to the other side and crash against the lower shroud, perhaps breaking the shroud or itself. The boat may

also broach if the seas are running fairly high. Again, if something breaks or the boat broaches, the vessel is no longer racing. In fact, it is instantaneously in danger of losing its rig and getting rolled over. It will be out of the race.

On *Project Amazon*, since there are no wires, the booms can set way forward of abeam, wing-and-wing. Reidl can sail 90° by the lee if he wants to without jibing. If the boat starts to round up because of a hit by a wave or gust, the sails will naturally pull the bow back downwind. And even if Reidl does jibe, what happens if the boom gets away from him? Nothing, because there is nothing to hit—neither wires nor boom can

Project Amazon's masts have an entasis taper (described in the text) that is stronger, stiffer, and more aesthetically pleasing than a straight taper (see profile and end view). The load on each mast is represented by the bending moment distribution curve. Maximum theoretical bending moment load on the mast is 40% of its breaking strength.

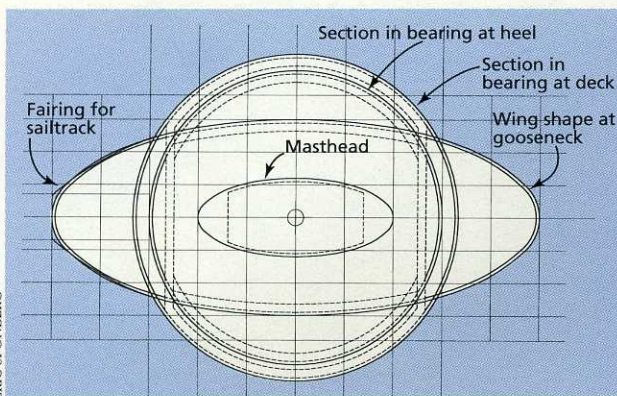
break. The sails will stop in a luff position all by themselves. It is very unlikely that the boat will broach. I won't say it will never broach, but because of the design of both the rig and the hull—a shallow V-bottom that tracks well—its chances of broaching, and, therefore, not racing, are exceedingly slim.

Mast Design

Boats racing in the Around Alone race are divided into two "open class" categories, and the design rule is very simple. Class I boats are between 50' and 60' LOA, while Class II boats are between 40' and 50' LOA. That's all the hull measurement required. Some safety rules govern the number and placement of bulkheads, the minimum righting arm, and the maximum allowable angle that the boat can heel with water ballast full one side, or with a swinging keel, such as *Project Amazon's*, swung full to one side. The only rig restriction is that sails and their spars cannot extend beyond the ends of the boat more than a total of 10% of the boat's length. Every other design aspect of the boat is totally "open," hence the name.

The design of a free-standing mast is fairly straightforward. Whereas a stayed mast carries its load in compression, owing to the pull of the rigging wires, a free-standing mast always bends. The pressure of the wind on the sails causes a bending moment load on the mast which is zero at the top, maximum at the deck, and zero again at the heel (see **diagram left**). The maximum possible bending moment the free-standing mast can carry is equal to the maximum righting moment of the boat itself, simply due to the physics of a boat floating free on the water. No matter how much sail area the mast carries, or how strong the wind is blowing, the load on the mast in normal sailing conditions can never exceed this value.

To determine the design bending moment for the mast, I multiply the boat's righting moment by a factor of safety of 2.5. In composite poles that bend, the reinforcing fibers generally start to break at about 50% of the breaking load of the pole, causing permanent loss of strength. Therefore, setting the design breaking load of the mast at 2.5 times the maximum righting moment of the boat means



A cross-sectional bird's-eye view of Project Amazon's masts. The dashed lines show the shape of the carbon fiber spar, which is approximately rectangular inside the elliptical wing shape, but round where it fits inside large roller bearings at the deck and heel.

that the actual bending load will never be higher than 40% of the design breaking load, and most sailing loads are much lower than this; the fibers won't break.

A free-standing mast must be as slender as possible consistent with strength, stiffness, and weight. Generally, larger cross-sectional shapes make lighter masts because the wall thickness gets thinner with increasing section size. There comes a point, however, where section size is too big and wall thickness too thin, making the mast too delicate to handle or subject to premature buckling. Therefore, every mast has an optimum size for each

boat design, and wall thicknesses should be between $\frac{1}{8}$ " and $\frac{1}{2}$ " in most instances.

Round, elliptical, and square mast-sections are the easiest to engineer because the geometric properties needed to calculate strength and stiffness (area, wall thickness, moment of inertia, and section modulus) can all be defined by simple equations. Airfoil-shaped sections, on the other hand, require some rather complicated calculus to determine strength and stiffness properties, so the engineering is much more difficult. On *Project Amazon*, I used an ellipse for the overall mast cross-sectional shape, which is

comprised of lightweight fiberglass leading- and trailing-edges bonded onto an approximately rectangular carbon fiber structural spar (see **diagram left**). At deck level and below, the spar changes from rectangular to round so that it can rotate in bearings.

An ellipse makes a great airfoil shape because the leading-edge curvature is exactly what airflow likes to see to remain smooth and attached to the surface. If the leading edge is too round, as in a cylindrical section, airflow near the mast surface has to change direction so quickly that it can separate from the mast, thereby reducing lift and increasing drag. A round section may be easy to engineer and build, but it is not the best shape aerodynamically.

The ellipse should not be too narrow, because narrower sections are more likely to buckle sideways. The mast should bend nicely both to the side as well as aft, and a fatter section does this well. Also, a fatter section has a wider window for angle of attack to the wind than does a narrower section. When a boat sails over waves, the mast moves all over the place, and the angle of attack on the rig—and thus lift—changes constantly.



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If the angle of attack gets too high, the airflow separates off the leading edge of the mast. This is called stall, and lift suddenly goes to zero. A narrow section stalls at a lower angle of attack than a fatter section. Also, a fatter mast is much more forgiving and easier to trim than a narrow mast. On *Project Amazon*, the elliptical mast section shape has a 2.5:1 chord-to-thickness ratio. This is about the maximum ratio I would ever go. Some of my designs are at about 2.0:1, and the lowest I have gone is 1.25:1, almost round.

Mast taper is important, too. I never use a straight taper above the deck because it is not structurally advantageous, and such masts look too spindly when viewed from deck level. I favor a modified entasis (en'-ta-sis) taper, employed by the Greeks over 2,500 years ago to build the columns of their temples to the gods—fatter in the middle and tapering nonlinearly toward the ends, which prevents the illusion of concavity as produced by a perfectly straight column. Columns must also be stronger in the middle, so entasis taper also builds in strength.

The strongest part of a mast, however, and, therefore, the biggest cross-section, has to be at deck level. After a straight

length from the deck up past the gooseneck, the mast tapers according to a mathematical formula, gently at first, then more so up toward the top. This modified entasis makes the mast fatter in the mid-height region than would a straight taper, and fatter is stronger, stiffer, and lighter.

Also, viewed from below, an entasis taper is more aesthetically pleasing than a straight taper, the very reason the Greeks used it. Below deck level, the lower end of the mast is slightly smaller so that it will fit through the deck opening, and I use a straight taper from deck to heel for ease of construction and because such aesthetic considerations here are not important.

Mast Construction

Project Amazon's masts were built by Composite Engineering in Concord, Massachusetts (see PBB No. 41, page 28,



ERIC SPONBERG

At Composite Engineering, the masts' pre-preg carbon-honeycomb preforms have been autoclave-cured, glued together, and braided with more carbon prior to adding fore-and-aft fairing strips.

for a feature article on this shop), and shipped to the boat, which was being built in Cape Town, South Africa. Composite Engineering (CE) opted for a combination of pre-preg molding with an autoclave, and wet-layup cured under vacuum bag and heat. Two different processes were involved because the masts are 85' long, and the autoclave is only 60' long.



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Smaller sections were molded in the autoclave, but it then served simply as an oven for the larger sections.

The masts are symmetrical front to back and side to side, so we only needed female forms to mold one half of a single mast. CE made 6'-long sheet-metal molds for one side of the structural spar and one leading/trailing edge shape. In the spar molds, the CE crew laid up thin, honeycomb-cored, carbon fiber shells, which were all bonded together to form the inside surface of the full structural

spar. CE then braided the remaining carbon fiber laminate over this spar preform. After completing the structural spars, CE molded the leading and trailing edges and bonded them onto the mast, and then finished off with the various molded fittings.

While this may seem an efficient way to go, we were under a time constraint to meet the ship that was to take the masts to Africa. In fact, each mold had to be cycled four times, twice each mast for both masts. When you figure in the lam-

inating and curing times for each part, coupled with their assembly and later-stage molding, the manufacturing schedule became quite complicated. In the end, to meet the shipping schedule, the masts left CE without the final finish on them. Then, as seems to often happen, the ship was two weeks late leaving the country, and the spars sat in storage all that time. To add insult to injury, while the ship was anchored in Lagos, Nigeria, it was rammed and holed by another ship, and Reidl had to fly there immediately to rescue his precious cargo from sinking into the murky depths. But that's another story.

Project Amazon's rig gives it tactical advantages that the other, conventionally rigged boats do not have. Those boats rely on very light weight to achieve impressive broad-reaching speed. But for at least half the race, that speed is in the wrong direction, because to achieve those high speeds, the boats have to tack downwind at roughly 40° either side of the course line. Speed on the course line is actually 24% less than boat speed—15 knots on track becomes 11½ knots on the course. The conventional boats are also difficult to control because their round hull shapes are very shallow, and their exceedingly thin keels have minuscule projected area. As a result, they have comparatively poor directional stability and control. Speed without control is worthless; the propensity to crash is high. *Project Amazon's* rig allows it to sail deep downwind at high speed—15 knots easily at 180° to true wind—under full control. The hull and keel shapes want to stay on course, and the rig is inherently stable, so the propensity to crash is very low.

I hope that Reidl's performance with *Project Amazon* will be respectable enough that the rest of the sailing world will take notice. The philosophy of the design applies directly to the cruising-boat market and even to racing boats. We have a continuing history of development of free-standing rigs since the beginning of Freedom Yachts 22 years ago. The sidebar beginning on page 46 traces some significant boats and their stories, including more than a few important race victories. Without rigging wires, the cost, complication, and maintenance of a boat go down, and the ease of sailing and its enjoyment go up.

And enjoyment on the water is what sailing is really all about. **PBB**

About the Author: Newport, Rhode Island-based naval architect Eric Sponberg is a frequent contributor to *Professional BoatBuilder*. His most recent design is a new 42' Globetrotter cat-ketch.



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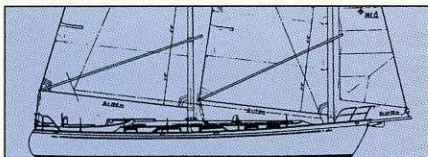
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Continues from page 46



COURTESY OF TANTON, INC.

TANTON 45

Tanton Inc.

This is the big sister to the Offshore 33 and 43, which were also Tanton designs but are no longer in production. Five Tanton 45s have been built to date, and they are available through Tanton on a semi-custom basis, complete with carbon fiber masts and wishbone booms. The design's offshore pedigree and capability are based on the success of the 48 Tanton 43s that have cruised over all the oceans and around the world. The 43s have won a 1,000-mile Bermuda Race and have sailed the Jimmy Cornell Transatlantic race a number of times. One couple, each over 65 years old, set a cruising speed record for Cape Town to Rio. "The cat-ketch rig is the heart of this boat in terms of stability, performance, and ease of handling," Tanton says. He admits that the public has a tough time understanding, and getting used to, this rig. With a chuckle, Tanton speculates that he should "put on fake spreaders and shrouds, just to make it look normal, then gradually eliminate them."

ERIC SPONBERG



CORROBOREE

Sponberg 35

Owners: Bill and Marilyn McBain, Gibraltar, Michigan.

Inspired by the early Freedom sloops, this wood-and-epoxy sloop was built in New Zealand in 1986-87 while the McBains were stationed in Australia. The carbon fiber mast was built in New Hampshire by PLP Composite Technologies Inc. A boatyard fire in 1995 destroyed the original mast, which was otherwise in nearly pristine condition after 10 years of Great Lakes sailing. The McBains say that *Corroborree* generally outperforms most other cruising boats of similar size. *Corroborree* was named by *Sail* magazine as one of the 100 greatest sailing yachts of North America in 1993.

RODGER MARTIN



GRAY WOLF

Rodger Martin custom 40

Owner: Eric Urbahn.

Built in 1996 by Lyman-Morse (Thomaston, Maine) in wood and epoxy, *Gray Wolf* also has water ballast. The free-standing sloop-rigged mast was built by Composite Engineering. *Gray Wolf* easily points 30° to the apparent wind, and has clocked 12.6 knots on a spinnaker reach in 16 knots of breeze. *Gray Wolf* swept the 1996 Eggemoggin Reach Regatta in Maine with line honors, first in class, and first on handicap, beating the 90' *Whitehawk* boat-for-boat by over three minutes, and the 55' *Vortex* by almost a half hour. *Gray Wolf* is now known in Maine as "that friggin' black boat." *Gray Wolf* also won the 1996 Opera House Cup, Class A.

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SALT

Craig Walters 49

Owner: Jon Roop, Beaufort, North Carolina.

This very attractive Russ Russell-built custom wood ketch "looks normal," but her masts are free-standing, constructed of wood-and-epoxy and S-2 glass to a Sponberg design. Check stays on the mainmast keep the headstay tight. S-2 glass is as strong as carbon fiber but not as stiff, and the mainmast was found to be too bendy when *Salt* was launched in 1991.

Diamond wires and spreaders, attached only to the mainmast itself, cured the problem. Roop charts *Salt* in the Caribbean.



MAGIE NOIRE

Tanton 70

Owner: Christian Quoniam, Fort de France, Martinique.

Launched in 1986, this lightweight cat-ketch charts in the French Antilles, and it has crossed the Atlantic 21 times. Naval architect Yves-Marie Tanton has long been a pioneer, promoter, designer, and builder of free-standing rigs, as in his Tanton 33, 43, and 45 (see earlier in this list). *Magie Noir's* wingmast rig is noteworthy for its D-section mast shape and double-ply sails that are cambered like airplane wings on either tack.

A final note about the virtues of the free-standing cat-ketch rig. Cat-ketches point well, they are particularly fast off the wind, they are easy to handle in all conditions, and they've proven themselves time and again to be capable of withstanding the demands of offshore racing and transoceanic cruising.

—E.S.

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